

Q1 - 25 July - Shift 1

Let $a_1 = b_1 = 1$, $a_n = a_{n-1} + 2$ and $b_n = a_n + b_{n-1}$ for every natural number $n \geq 2$. Then $\sum_{n=1}^{15} a_n \cdot b_n$ is equal to _____.

Space for your notes:

Q2 - 25 July - Shift 2

The sum $\sum_{n=1}^{21} \frac{3}{(4n-1)(4n+3)}$ is equal to

Space for your notes:

(A) $\frac{7}{87}$

(B) $\frac{7}{29}$

(C) $\frac{14}{87}$

(D) $\frac{21}{29}$

Q3 - 26 July - Shift 1

Consider two G.Ps. $2, 2^2, 2^3, \dots$ and $4, 4^2, 4^3, \dots$ of 60 and n terms respectively. If the geometric mean of all the $60 + n$ terms is $(2)^{\frac{225}{8}}$, then $\sum_{k=1}^n k(n-k)$ is equal to :

Space for your notes:

(A) 560

(B) 1540

(C) 1330

(D) 2600

Q4 - 26 July - Shift 1

Questions

MathonGo

The series of positive multiples of 3 is divided into sets : {3}, {6, 9, 12}, {15, 18, 21, 24, 27},... Then the sum of the elements in the 11th set is equal to _____,

*Space for your notes:***Q5 - 26 July - Shift 2**

If $\sum_{k=1}^{10} \frac{k}{k^4 + k^2 + 1} = \frac{m}{n}$, where m and n are co-prime, then m + n is equal to

*Space for your notes:***Q6 - 26 July - Shift 2**

Different A.P.'s are constructed with the first term 100, the last term 199, And integral common differences. The sum of the common differences of all such, A.P's having at least 3 terms and at most 33 terms is.

*Space for your notes:***Q7 - 27 July - Shift 1**

Suppose $a_1, a_2, \dots, a_n, \dots$ be an arithmetic progression of natural numbers. If the ratio of the sum of the first five terms of the sum of first nine terms of the progression is 5 : 17 and $110 < a_{15} < 120$, then the sum of the first ten terms of the progression is equal to -

Space for your notes:

- (A) 290 (B) 380
(C) 460 (D) 510

Q8 - 27 July - Shift 1

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Questions

MathonGo

Let $f(x) = 2x^2 - x - 1$ and

Space for your notes:

$S = \{n \in \mathbb{Z} : |f(n)| \leq 800\}$. Then, the value of

$\sum_{n \in S} f(n)$ is equal to _____.

Q9 - 27 July - Shift 2

Let the sum of an infinite G.P., whose first term is a and the common ratio is r , be 5. Let the sum of its first five terms be $\frac{98}{25}$. Then the sum of the first

Space for your notes:

21 terms of an AP, whose first term is $10ar$, n^{th} term is a_n and the common difference is $10ar^2$, is equal to :

(A) $21 a_{11}$ (B) $22 a_{11}$

(C) $15 a_{16}$ (D) $14 a_{16}$

Q10 - 27 July - Shift 2

$$\frac{2^3 - 1^3}{1 \times 7} + \frac{4^3 - 3^3 + 2^3 - 1^3}{2 \times 11} + \frac{6^3 - 5^3 + 4^3 - 3^3 + 2^3 - 1^3}{3 \times 15} + \dots + \frac{30^3 - 29^3 + 28^3 - 27^3 + \dots + 2^3 - 1^3}{15 \times 63}$$

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is equal to _____.

Q11 - 28 July - Shift 1

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Questions

MathonGo

Consider the sequence $a_1, a_2, a_3, \dots$ such that

$$a_1 = 1, a_2 = 2 \text{ and } a_{n+2} = \frac{2}{a_{n+1}} + a_n \text{ for } n = 1, 2, 3, \dots$$

If
$$\left(\frac{a_1 + \frac{1}{a_2}}{a_3} \right) \cdot \left(\frac{a_2 + \frac{1}{a_3}}{a_4} \right) \cdot \left(\frac{a_3 + \frac{1}{a_4}}{a_5} \right) \dots \left(\frac{a_{30} + \frac{1}{a_{31}}}{a_{32}} \right) = 2^\alpha \left({}^{61}C_{31} \right),$$

then α is equal to :

(A) -30

(B) -31

(C) -60

(D) -61

Space for your notes:

Q12 - 28 July - Shift 1

For $p, q \in \mathbb{R}$, consider the real valued function

$f(x) = (x - p)^2 - q$, $x \in \mathbb{R}$ and $q > 0$. Let a_1, a_2, a_3 and a_4 be in an arithmetic progression with mean p and positive common difference. If $|f(a_i)| = 500$ for all $i = 1, 2, 3, 4$, then the absolute difference between the roots of $f(x) = 0$ is

Space for your notes:

Q13 - 28 July - Shift 1

Let $x_1, x_2, x_3, \dots, x_{20}$ be in geometric progression

with $x_1 = 3$ and the common ratio $\frac{1}{2}$. A new data

is constructed replacing each x_i by $(x_i - i)^2$. If $\bar{x}$ is the mean of new data, then the greatest integer less than or equal to $\bar{x}$ is _____.

Space for your notes:

Q14 - 28 July - Shift 2

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Questions

MathonGo

If $\frac{6}{3^{12}} + \frac{10}{3^{11}} + \frac{20}{3^{10}} + \frac{40}{3^9} + \dots + \frac{10240}{3} = 2^n \cdot m$,

where m is odd, then m.n is equal to _____

Space for your notes:

Q15 - 29 July - Shift 1

If $\frac{1}{(20-a)(40-a)} + \frac{1}{(40-a)(60-a)} + \dots + \frac{1}{(180-a)(200-a)} = \frac{1}{256}$, then the maximum value of a is :

- (A) 198 (B) 202
(C) 212 (D) 218

Space for your notes:

Q16 - 29 July - Shift 1

Let $a_1, a_2, a_3, \dots$ be an A.P. If $\sum_{r=1}^{\infty} \frac{a_r}{2^r} = 4$, then $4a_2$ is equal to _____.

Space for your notes:

Q17 - 29 July - Shift 1

If $\frac{1}{2 \times 3 \times 4} + \frac{1}{3 \times 4 \times 5} + \frac{1}{4 \times 5 \times 6} + \dots + \frac{1}{100 \times 101 \times 102} = \frac{k}{101}$, then $34k$ is equal to _____.

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Q18 - 29 July - Shift 2

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Questions

MathonGo

Let $\{a_n\}_{n=0}^{\infty}$ be a sequence such that $a_0 = a_1 = 0$ and

Space for your notes:

$$a_{n+2} = 3a_{n+1} - 2a_n, \forall n \geq 0.$$

Then $a_{25} a_{23} - 2 a_{25} a_{22} - 2 a_{23} a_{24} + 4 a_{22} a_{24}$ is equal to:

- (A) 483 (B) 528 (C) 575 (D) 624

Q19 - 29 July - Shift 2

$\sum_{r=1}^{20} (r^2 + 1)(r!)$ is equal to:

Space for your notes:

- (A) $22! - 21!$ (B) $22! - 2(21!)$
(C) $21! - 2(20!)$ (D) $21! - 20!$

Answer Key**Q1** (27560)**Q2** (B)**Q3** (C)**Q4** (6993)**Q5** (166)**Q6** (53)**Q7** (B)**Q8** (10620)**Q9** (A)**Q10** (120)**Q11** (C)**Q12** (50)**Q13** (142)**Q14** (12)**Q15** (C)**Q16** (16)**Q17** (286)**Q18** (B)**Q19** (B)

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Q1 (27560)					
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$$a_1 = b_1 = 1$$

$$a_2 = a_1 + 2 = 3$$

$$a_3 = a_2 + 2 = 5$$

$$a_4 = a_3 + 2 = 7$$

$$\Rightarrow a_n = 2n - 1$$

$$b_2 = a_1 + b_1 = 4$$

$$b_3 = a_3 + b_2 = 9$$

$$b_4 = a_4 + b_3 = 16$$

$$b_n = n^2$$

$$\sum_{n=1}^{15} a_n b_n$$

$$\sum_{n=1}^{15} (2n-1)n^2$$

$$\sum_{n=1}^{15} (2n^3 - n^2)$$

$$= 2 \frac{n^2(n+1)^2}{4} - \frac{n(n+1)(2n+1)}{6}$$

$$\text{Put } n = 15$$

$$= \frac{2 \times 225 \times 16 \times 16}{4} - \frac{15 \times 16 \times 31}{6}$$

Q2 (B)

(B)

Q3 (C)

$$\left((2^1 2^2 \dots 2^{60}) (4^1 4^2 \dots 4^n) \right)^{\frac{1}{60+n}} = 2^{\frac{225}{8}}$$

$$\left(2^{30 \times 61} 4^{\frac{n(n+1)}{2}} \right)^{\frac{1}{60+n}} = 2^{\frac{225}{8}}$$

$$2^{1830+n^2+n} = 2^{\frac{(225)(60+n)}{8}}$$

$$= 8n^2 - 217n + 1140 = 0$$

$$n = 20, \frac{57}{8}$$

$$\sum_{k=1}^n nk - k^2 = \frac{n^2(n+1)}{2} - \frac{n(n+1)(2n+1)}{6}$$

$$= 1330$$

Q4 (6993)

$$S_{11} = 3[101 + 102 + \dots + 121]$$

$$= \frac{3}{2}(222) \times 21 = 6993$$

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Q5 (166)

$$\sum_{k=1}^{10} \frac{k}{k^4 + k^2 + 1} = \frac{m}{n}$$

$$\Rightarrow \frac{1}{2} \sum_{k=1}^{10} \frac{(k^2 + k + 1) - (k^2 - k + 1)}{(k^2 + k + 1)(k^2 - k + 1)}$$

$$\Rightarrow \frac{1}{2} \left(\sum_{k=1}^{10} \left(\frac{1}{(k^2 - k + 1)} - \frac{1}{k^2 + k + 1} \right) \right)$$

$$\Rightarrow \frac{55}{111} = \frac{m}{n}$$

$$m + n = 166$$

Q6 (53)

$$1^{\text{st}} \text{ term} = 100 = a$$

$$\text{Last term} = 199 = \ell$$

If 3 term

$$a, a + d, a + 2d$$

$$a_n = \ell = a + (n - 1)d$$

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$n \rightarrow$ number of terms

$$n=3, d_1 = \frac{199-100}{2}$$

$$= \frac{99}{2} \notin I$$

$$n=4, d_2 = \frac{99}{3} = 33 \in I$$

$$n=10, d_3 = \frac{99}{9} = 11 \in I$$

$$n=12, d_4 = \frac{99}{11} = 9 \in I$$

$$\therefore \sum d_i = 33 + 11 + 9 = 53$$

Q7 (B)

$$\frac{S_5}{S_9} = \frac{5}{17} \Rightarrow \frac{\frac{5}{2}(2a + 4d)}{\frac{9}{2}(2a + 8d)} = \frac{5}{17}$$

$$\Rightarrow d = 4a$$

$$a_{15} = a + 14d = 57a$$

$$\text{Now, } 110 < a_{15} < 120$$

$$\Rightarrow 110 < 57a < 120$$

$$\Rightarrow a = 2 \therefore d = 8$$

$$S_{10} = \frac{10}{2}(2 \times 2 + 9 \times 8) = 380$$

Q8 (10620)

$$f(x) = 2x^2 - x - 1$$

$$|f(x)| \leq 800$$

$$2n^2 - n - 801 \leq 0$$

$$n^2 - \frac{1}{2}n - \frac{801}{2} \leq 0$$

$$\left(n - \frac{1}{4}\right)^2 - \frac{801}{2} - \frac{1}{16} \leq 0$$

$$\left(n - \frac{1}{4}\right)^2 - \frac{6409}{16} \leq 0$$

$$\left(n - \frac{1}{4} - \frac{\sqrt{6409}}{4}\right)\left(n - \frac{1}{4} + \frac{\sqrt{6409}}{16}\right) \leq 0$$

$$\frac{1 - \sqrt{6409}}{4} \leq n \leq \frac{1 + \sqrt{6409}}{4}$$

$$n = \{-19, -18, -17, \dots, 0, 1, 2, \dots, 20\}$$

$$\sum_{n \in S} f(x) = \sum (2x^2 - x - 1)$$

$$= 2[19^2 + 18^2 + \dots + 1^2 + 1^2 + 2^2 + \dots + 19^2 + 20^2]$$

$$= 4[1^2 + 2^2 + \dots + 19^2] + 2[20^2] - 20 - 40$$

$$= \frac{4 \times 19 \times 20 \times (2 \times 19 + 1)}{6} + 2 \times 400 - 60$$

$$= \frac{4 \times 19 \times 20 \times 39}{6} + 800 - 60 - 9880 + 800 - 60$$

$$= 10620$$

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Q9 (A)

$$S_{21} = \frac{21}{2} [20ar + 20 \cdot 10 ar^2]$$

$$= 21 [10ar + 100ar^2]$$

$$= 21 \cdot a_{11}$$

Q10 (120)

$$T_n = \frac{2 \sum_{r=1}^n (2r)^3 - \left(\sum_{r=1}^{2n} r^3 \right)}{n(4n+3)}$$

$$\Rightarrow T_n = n$$

$$\text{So, } \sum_{n=1}^{15} T_n = 120$$

Q11 (C)

$$a_{n+2} a_{n+1} - a_{n+1} \cdot a_n = 2$$

Series will satisfy

$$a_1 a_2, a_2 a_3, a_3 a_4, a_4 a_5, \dots$$

$$\begin{aligned}
 \frac{a_n + \frac{1}{a_{n+1}}}{a_{n+2}} &= \frac{a_{n+2} - \frac{1}{a_{n+1}}}{a_{n+2}} \\
 &= 1 - \frac{1}{a_{n+1}a_{n+2}} \\
 &= 1 - \frac{1}{2(r+1)} \\
 &= \frac{2r+1}{2(r+1)}
 \end{aligned}$$

Now proof is given by

$$\begin{aligned}
 &= \prod_{r=1}^{30} \frac{(2r+1)}{2(r+1)} \\
 &= \frac{(1 \cdot 3 \cdot 5 \cdot \dots \cdot 61)}{2^{30} \cdot (2 \cdot 3 \cdot \dots \cdot 31)} \\
 &\Rightarrow \frac{(1 \cdot 3 \cdot 5 \cdot \dots \cdot 61)}{\underline{31} \cdot 2^{30}} \times \frac{2^{30} \times \underline{30}}{2^{30} \times \underline{30}}
 \end{aligned}$$

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$$\alpha = -60$$

Q12 (50)

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$$f(x) = 0 \Rightarrow (x - p)^2 - q = 0.$$

Roots are $p + \sqrt{q}$, $p - \sqrt{q}$ absolute difference between roots $2\sqrt{q}$.

$$\text{Now, } |f(a_1)| = 500$$

Let a_1, a_2, a_3, a_4 are a_1 $a + d, a + 2d, a + 3d$

$$|f(a_4)| = 500$$

$$|(a_1 - p)^2 - q| = 500$$

$$\Rightarrow (a_1 - p)^2 - q = 500$$

$$\Rightarrow \frac{9}{4}d^2 - q = 500 \quad \text{_____ (1)}$$

$$\text{and } |f(a_1)|^2 = |f(a_2)|^2$$

$$((a_1 - p)^2 - q)^2 = ((a_2 - p)^2 - q)^2$$

$$\Rightarrow ((a_1 - p)^2 - (a_2 - p)^2) ((a_1 - p)^2 - q + (a_2 - p)^2 - q) = 0$$

$$\Rightarrow \frac{9}{4}d^2 - q + \frac{d^2}{4} - q = 0$$

$$2q = \frac{10d^2}{4} \Rightarrow q = \frac{5d^2}{4}$$

$$\Rightarrow d^2 = \frac{4q}{5}$$

$$\text{From equation (1)} \quad \frac{9}{4} \cdot \frac{4q}{5} - q = 500$$

$$\frac{4q}{5} = 500$$

$$\frac{4q}{5} = 500$$

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Hints and Solutions

MathonGo

Q13 (142)

$$\sum_{i=0}^{20} x_i = \frac{3 \left(1 - \left(\frac{1}{2} \right)^{20} \right)}{1 - \frac{1}{2}} = 6 \left(1 - \frac{1}{2^{20}} \right)$$

$$= \sum_{i=1}^{20} (x_{i-1})^2$$

$$= \sum_{i=1}^{20} (x_i)^2 + (i)^2 - 2x_i i$$

$$\text{Now } \sum_{i=1}^{20} (x_i)^2 = \frac{9 \left(1 - \left(\frac{1}{4} \right)^{20} \right)}{1 - \frac{1}{4}} = 12 \left(1 - \frac{1}{2^{40}} \right)$$

$$\sum_{i=1}^{20} i^2 = \frac{1}{6} \times 20 \times 21 \times 41 = 2870$$

$$\sum_{i=1}^{20} x_i i = S = 3 + 2.3 \frac{1}{2} + 3.3 \frac{1}{2^2} + 4.3 \frac{1}{2^3} + \dots \text{AGP}$$

$$= 6 \left(2 - \frac{22}{2^{20}} \right)$$

$$\bar{x} = \frac{12 - \frac{12}{2^{40}} + 2870 - 12 \left(2 - \frac{22}{2^{20}} \right)}{20}$$

$$\bar{x} = \frac{2858}{20} + \left(\frac{-12}{2^{40}} + \frac{22}{2^{20}} \right) \times \frac{1}{20}$$

$$[\bar{x}] = 142$$

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Q14 (12)

$$\frac{6}{3^{12}} + 10 \left(\frac{1}{3^{11}} + \frac{2}{3^{10}} + \frac{2^2}{3^9} + \frac{2^3}{3^8} + \dots + \frac{2^{10}}{3} \right)$$

$$\frac{6}{3^{12}} + \frac{10}{3^{11}} \left(\frac{6^{11} - 1}{6 - 1} \right)$$

$$= 2^{12} \cdot 1; m.n = 12$$

Q15 (C)

By splitting

$$\frac{1}{20} \left[\left(\frac{1}{20-a} - \frac{1}{40-a} \right) + \left(\frac{1}{40-a} - \frac{1}{60-a} \right) \right.$$

$$\left. + \dots + \left(\frac{1}{180-a} - \frac{1}{200-a} \right) \right]$$

$$\Rightarrow \frac{1}{20} \left(\frac{1}{20-a} - \frac{1}{200-a} \right) = \frac{1}{256}$$

$$(20-a)(200-a) = 256 \times 9$$

$$a^2 - 220a + 1696 = 0$$

$$a = 8, 212$$

Hence maximum value of a is 212.

Q16 (16)

$$S = \frac{a_1}{2} + \frac{a_2}{2^2} + \frac{a_3}{2^3} + \dots$$

$$\frac{S}{2} = \frac{a_1}{2^2} + \frac{a_2}{2^3} + \dots$$

$$\frac{S}{2} = \frac{a_1}{2} + d \left(\frac{1}{2^2} + \frac{1}{2^3} + \dots \right)$$

$$\frac{S}{2} = \frac{a_1}{2} + d \left(\frac{\frac{1}{4}}{1 - \frac{1}{2}} \right)$$

$$\therefore S = a_1 + d = a_2 = 4$$

$$\text{Or } 4a_2 = 16$$

Q17 (286)

$$\frac{1}{2.3.4} + \frac{1}{3.4.5} + \dots + \frac{1}{100.101.102} = \frac{k}{101}$$

$$\frac{4-2}{2.3.4} + \frac{5-3}{3.4.5} + \dots + \frac{102-100}{100.101.102} = \frac{2k}{101}$$

$$\frac{1}{2.3} - \frac{1}{3.4} + \frac{1}{3.4} - \frac{1}{4.5} + \dots + \frac{1}{100.101} - \frac{1}{101.102} = \frac{2k}{101}$$

$$\frac{1}{2.3} - \frac{1}{101.102} = \frac{2k}{101}$$

$$\therefore 2k = \frac{101}{6} - \frac{1}{102}$$

$$\therefore 34k = 286$$

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Q18 (B)

$$a_0 = 0, a_1 = 0$$

$$a_{n+2} = 3a_{n+1} - 2a_n : n \geq 0$$

$$a_{n+2} - a_{n+1} = 2(a_{n+1} - a_n) + 1$$

$$n = 0 \quad a_2 - a_1 = 2(a_1 - a_0) + 1$$

$$n = 1 \quad a_3 - a_2 = 2(a_2 - a_1) + 1$$

$$n = 2 \quad a_4 - a_3 = 2(a_3 - a_2) + 1$$

$$n = n \quad a_{n+2} - a_{n+1} = 2(a_{n+1} - a_n) + 1$$

$$(a_{n+2} - a_1) - 2(a_{n+1} - a_0) - (n+1) = 0$$

$$a_{n+2} = 2a_{n+1} + (n+1)$$

$$n \rightarrow n-2$$

$$a_n - 2a_{n-1} = n - 1$$

$$\text{Now } a_{25}a_{23} - 2a_{25}a_{22} - 2a_{23}a_{24} + 4a_{22}a_{24}$$

$$= (a_{25} - 2a_{24})(a_{23} - 2a_{22}) = (24)(22) = 528$$

Q19 (B)

$$\begin{aligned}
 & \sum_{x=1}^{20} (r^2 + 1)r! \\
 & \sum_{x=1}^{20} ((r+1)^2 - 2r)r! \\
 & \sum_{x=1}^{20} ((r+1)(r+1)! - r.r!) - \sum_{r=1}^{20} r.r! \\
 & \sum_{x=1}^{20} ((r+1)(r+1)! - r.r!) - \sum_{r=1}^{20} ((r+1)! - r!) \\
 & = (21 \cdot 21! - (21-1)) \\
 & = 20 \cdot 21! = 22! - 2 \cdot 21!
 \end{aligned}$$

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